

Grant Yang

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(408) 568-4549

SKILLS

Languages: Lean4, Python, C/C++, Rust, Java, TypeScript.

Technology: PyTorch, OpenGL, Git, Linux/Unix, Networking.

EXPERIENCE

UW Math AI Lab - Student Researcher (Spring 2026) [monogenic-exetensions](#)

- Worked with Prof. Bianca Viray to formalize select results from her paper [\[2503.07846\]](#) using Lean. Namely, that finite étale extensions of local rings are monogenic.
- Contributed code to Mathlib: [PR#36421](#) and [PR#37527](#).

Euler Circle - Teaching Assistant (Summer 2026) [Mathematics in Lean](#)

- Taught class on real analysis and Lean for advanced high school students.
- Held office hours/homework sessions and graded student work.

PERSONAL PROJECTS

UW Lean Hackathon - 1st Place [leomayer1/cat-rw](#), [uw2026leanhackathon.github.io](#)

- Won first prize for a tactic to perform substitutions, rewriting data using isomorphisms as if they were simple equalities. Useful for category-theoretic proofs.
- Automated the process of writing these tedious proofs.

ByteGPT [Online demo](#)

- Trained a tiny 20M-parameter transformer using PyTorch with a custom architecture.
- Hand-wrote a highly efficient C++ inference engine targeting WebAssembly.
 - * Runs in-browser at FP32 using only 300 MB for 4096 context, at over 40 tok/s.

C++ Chess Engine (No AI code!) [ROTARTSI82/chess](#)

- Wrote a multithreaded chess engine with magic bitboards and alpha-beta search.
- Optimized to search millions of positions per second, reaching over 1600 Lichess rating.

Pascal Compiler [ROTARTSI82/ATCS](#)

- Designed and built a compiler for a custom Pascal dialect targeting MIPS32 assembly.
- Demonstrated complex programs on a MIPS32 emulator.

EDUCATION

University of Washington Seattle, WA

B.S. in Electrical and Computer Engineering 2025 – 2029

- **Coursework:** MATH 208: Linear Algebra, MATH 327: Real Analysis, MATH 380: Computational Algebraic Geometry, STAT 390: Intro to Statistics.
- **GPA:** 3.99/4.0

Harker Upper School San Jose, CA

4.6/4.8 GPA 2021 – 2025